

Correlated Decryption Errors in DTRU: Revisiting Failure Rates and Failure Boosting for Lattice KEMs with Repetition-Coded Blocks

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Abstract

DTRU is an NTRU-based key encapsulation mechanism submitted to the Chinese NGCC post-quantum competition. Its designers bound the decryption failure rate (DFR) of every parameter set by modelling the 16 error coefficients of each DoubleE₈ decoding block as independent Gaussians and using the inscribed ball of the code’s Voronoi cell as a sufficient decoding condition. We show that both steps fail for the tricyclotomic ring $\mathbb{Z}_q[x]/(x^n - x^{n/2} + 1)$ and for the LPPNF ring $\mathbb{Z}_q[x]/(x^n - x - 1)$ used by six of the seven parameter sets. Ring multiplication couples coefficients i and $i + n/2$, the repetition code forces $w_i = w_{i+n/2}$, and each decoding block consists of exactly eight such pairs, so the block covariance has eight large and eight small eigenvalues. Replacing the ball by the exact maximum-likelihood failure region of Algorithm 2 and the independence assumption by the conditional block covariance, we estimate the true DFR of DTRU-648, -768, -1024, -Prime, -1536 and -2048 to lie between 2^{-85} and 2^{-123} , against claimed values of 2^{-147} to 2^{-204} . The model is validated against the real decoder on scaled-down instances of all three rings to within one bit at the median. In the failure-boosting cost model of D’Anvers et al., one decryption failure of DTRU-648 costs about 2^{85} decapsulations, far below the claimed 128-bit level; when the decapsulation oracle is capped at 2^{64} queries, an offline precomputation of 2^{22} – 2^{26} encapsulations suffices for DTRU-648, -768 and -Prime. DTRU-Light, defined over $x^n + 1$, has no such pairing and its DFR stays at the claimed order. No attack we describe is practical; the results invalidate the stated security levels of DTRU in the DFR metric and the methodology used to compute them.

1 Introduction

Lattice-based KEMs built from (Ring/Module-)LWE or NTRU decrypt correctly only with overwhelming probability. A small decryption failure rate (DFR) δ is harmless for correctness, but it enters the security of the Fujisaki–Okamoto transform through terms of the form $(q_H + q_D)\delta$, and, more concretely, an adversary who can trigger failures learns information about the secret key. D’Anvers, Vercauteren and Verbaauwhede [2] introduced *failure boosting*: honestly generate many ciphertexts offline, keep those whose failure probability is unusually high, and submit only those to the decapsulation oracle. D’Anvers, Rossi and Virdia [3] showed that once a first failure is found, subsequent failures are much cheaper (*directional failure boosting*), so the security target is that even one failure must cost 2^λ . D’Anvers and Batsleer [4] refined the cost accounting and applied it to Saber and Kyber.

Designers therefore need a reliable DFR estimate. The common practice is to compute the distribution of a single error coefficient exactly (by convolution) and then to treat the coefficients

that enter one decoding decision as independent. For schemes over $\mathbb{Z}_q[x]/(x^n + 1)$ with one bit per coefficient this is well studied [2, 4, 5]. Error-correcting codes change the picture: a decoding decision now depends on a block of coefficients, and any correlation inside the block changes the tail.

DTRU. DTRU [1] is an NTRU-based KEM submitted to the NGCC competition. Its main parameter sets live in the tricyclotomic ring $\mathcal{R}_q = \mathbb{Z}_q[x]/(x^n - x^{n/2} + 1)$ with $q = 3457$, a variant DTRU-Prime uses the LPPNF ring $\mathbb{Z}_q[x]/(x^{1087} - x - 1)$, and a lightweight set DTRU-Light uses $\mathbb{Z}_{769}[x]/(x^{512} + 1)$. Messages are encoded with DoubleEg: four bits become an $[8, 4, 4]$ extended Hamming codeword, which is then repeated by multiplication with $(1 + x^{n/2})$; the decoder works on 16-coefficient blocks consisting of the i -th octet and the octet at distance $n/2$. The specification bounds the DFR by (i) the inscribed ball $\|\text{Err}^{(i)}\|_2^2 < (q - 1)^2/2$ of the code and (ii) modelling $\|\text{Err}^{(i)}\|^2$ as $\sigma^2\chi^2(16)$, and reports failure rates from $2^{-143.9}$ (Light) to $2^{-204.1}$ (DTRU-2048).

Contributions.

1. **The failure event.** We derive the exact failure region of Algorithm 2. Because the two symbol values 0 and $(q + 1)/2$ are antipodal in $\mathbb{Z}_q$, block i is decoded incorrectly iff for some codeword difference with support D ($|D| \in \{8, 16\}$) $\sum_{j \in D} |\text{Err}_j| \geq |D|(q + 1)/4$. This is a union of half-spaces tangent to the inscribed ball, so the ball is a strict over-approximation.
2. **The correlation.** In $x^n - x^{n/2} + 1$ every cross term of a ring product pairs indices at distance exactly $n/2$, and the repetition code makes $w_i = w_{i+n/2}$. Each decoding block is made of eight such pairs. The conditional covariance of a block, given the ciphertext, has eight eigenvalues around $0.3\times$ and eight around $1.6\text{--}2.1\times$ the marginal variance. The failing directions are the large ones.
3. **A validated estimator.** We compute the Gaussian half-space union per block given the ciphertext randomness, correct the dominant terms with the exact centred-binomial cumulant generating function (Lugannani–Rice), and validate against the real key generation, encryption and decoder on scaled-down instances of all three rings: medians agree to within $0.3\text{--}0.8$ bits, second moments to $1\text{--}3\%$.
4. **Corrected DFRs.** For 200 honest ciphertexts per set, the median failure probabilities of DTRU-648/768/1024/Prime/1536/2048 are $2^{-97}, 2^{-112}, 2^{-118}, 2^{-115}, 2^{-111}, 2^{-123}$ and the averages $2^{-85}, 2^{-98}, 2^{-105}, 2^{-96}, 2^{-101}, 2^{-111}$; the claimed values are 2^{-147} to 2^{-204} . Re-running the specification’s own method with our variances reproduces their numbers to within ± 8 bits, so the gap is the model, not the variance. DTRU-Light has no pairing and its DFR is of the claimed order.
5. **Failure boosting costs.** Classical work for a first failure cannot go below $1/\mathbb{E}[\delta]$; we obtain $2^{85}\text{--}2^{111}$ for the six affected sets against nominal levels 128–512. Separating the offline precomputation from oracle queries as in [2], DTRU-648 already contains honest ciphertexts with failure probability above 2^{-80} among 200 samples, and a 2^{64} query cap is met with $2^{22}\text{--}2^{26}$ offline encapsulations for DTRU-648, -768 and -Prime.
6. **After the first failure.** A failure yields a known linear form in the secret with about 900 effective coordinates out of 4096 (DTRU-2048); failure directions of independent ciphertexts have cosine $0.19\text{--}0.27$, so a second failure costs about $2^{-72}\text{--}2^{-86}$ once the first is known, consistent with [3]. Full key recovery is not executed.

Every number in this paper is reproducible from a few hundred lines of Python; scripts and raw outputs accompany the paper.

Set	ring	p	n	q	q_2	$(k_g, k_{f'})$	(k_r, k_e)	claimed $\log_2 \delta$	level
Light	pow2	$1 - x^{n/2}$	512	769	2^8	(2, 1)	(1, 2)	-143.92	128
648	tri	2	648	3457	2^9	(9, 2)	(2, 2)	-146.79	128
768	tri	2	768	3457	2^{10}	(4, 3)	(4, 2)	-185.56	192
1024	tri	2	1024	3457	2^{10}	(5, 2)	(3, 2)	-190.48	256
Prime	lppnf	2	1087	2017	2^{10}	(2, 1)	(2, 1)	-180.72	256
1536	tri	2	1536	3457	2^{10}	(3, 2)	(2, 1)	-195.50	384
2048	tri	2	2048	3457	2^{10}	(5, 1)	(2, 1)	-204.10	512

Table 1: DTRU parameter sets. DTRU-648 targets 128-bit quantum security (its NTRU Core-SVP claim is 164/144).

2 Preliminaries

2.1 Notation

For a polynomial $a \in \mathcal{R} = \mathbb{Z}[x]/(\phi(x))$ of degree n we write a_j for its coefficients and $M_a \in \mathbb{Z}^{n \times n}$ for the matrix of multiplication by a , whose j -th column is $a \cdot x^j$; then $M_a b = ab$. For an index set I of size 16 we write $M_a[I]$ for the $16 \times n$ submatrix of rows I . $\mathcal{B}_\eta$ is the centred binomial distribution with variance $\eta/2$. $Q(z) = \Pr[N(0, 1) \geq z]$.

2.2 DTRU

Three rings are used:

$$\mathcal{R}^{\text{tri}} = \mathbb{Z}[x]/(x^n - x^{n/2} + 1), \quad \mathcal{R}^{\text{pow2}} = \mathbb{Z}[x]/(x^n + 1), \quad \mathcal{R}^{\text{lppnf}} = \mathbb{Z}[x]/(x^n - x - 1).$$

Key generation samples $f' \leftarrow \mathcal{B}_{k_{f'}}^n$, $g \leftarrow \mathcal{B}_{k_g}^n$, sets $f = pf' + 1$ with $p = 2$ (tricyclotomic, LPPNF) or $p = 1 - x^{n/2}$ (power-of-two), and publishes $h = g/f \bmod q$. Encryption samples $r \leftarrow \mathcal{B}_{k_r}^n$, $e \leftarrow \mathcal{B}_{k_e}^n$, encodes the message to $w \in \{0, 1\}^n$ and outputs $c = \lfloor \frac{q_2}{q}(hr + e + \frac{q+1}{2}w) \rfloor \bmod q_2$. Decryption decompresses $c' = \lfloor \frac{q}{q_2}c \rfloor$, computes $c'f$ and decodes. Writing ε for the compression noise and $u = e + \varepsilon$, the specification's Theorem 1 gives

$$c'f = \text{Err} + \frac{q+1}{2}w, \quad \text{Err} = \begin{cases} gr + f'(2u + w) + u, & p = 2, \\ gr + f'((1 - x^{n/2})u + w') + u, & p = 1 - x^{n/2}. \end{cases} \quad (1)$$

DoubleE₈. Each 4-bit chunk m_i is mapped to $u_i = m_i H \bmod 2 \in \{0, 1\}^8$ with H the generator of the $[8, 4, 4]$ extended Hamming code, placed at coefficients $8i, \dots, 8i + 7$ of w' , and $w = (1 + x^{\text{off}})w'$ with $\text{off} = n/2$. Hence $w_j = w_{j+n/2} = w'_j$. Algorithm 2 decodes block i from the 16 coefficients $t_{8i}, \dots, t_{8i+7}, t_{8i+n/2}, \dots, t_{8i+7+n/2}$ by computing, for each of the 16 codewords $\frac{q+1}{2}(u, u)$, the sum of squared centred distances $\|\cdot\|_{q,2}^2$ and returning the argmin. For $n = 1087$ (DTRU-Prime) the specification does not state the layout; we use $\lfloor n/16 \rfloor = 67$ blocks and $\text{off} = \lfloor n/2 \rfloor = 543$.

Parameters. Table 1 lists the seven sets (specification Table 1 and 2). The KEM is $\text{FO}_{\text{ID}(pk),m}^\chi$ with implicit rejection and $(K, \rho) = H(\text{ID}(pk), m)$; the encryption randomness is derived from ρ .

2.3 The specification's DFR bound

Section 2.3 of [1] argues that block i decodes correctly whenever

$$\|\text{Err}^{(i)}\|_2^2 = \|\text{Err}_i\|_{q,2}^2 + \|\text{Err}_{i+n/16}\|_{q,2}^2 < \frac{(q-1)^2}{2} =: T, \quad (2)$$

because the scaled code has minimum Euclidean distance $\sqrt{2}(q-1)$. It then assumes the 16 coefficients independent, so that $\|\text{Err}^{(i)}\|^2 \sim \sigma^2 \chi^2(16)$, and applies a union bound over $n/16$ blocks: $\delta \leq \frac{n}{16} \Pr[\chi^2(16) \geq T/\sigma^2]$. Section 2.4 adds that adversarial choice of the message can raise the failure rate by at most about 2^{20} .

2.4 Failure boosting

Following [2, 4], let $p(c)$ be the failure probability of an honestly generated ciphertext c over the secret key, and let $\delta = \mathbb{E}_c[p(c)]$. The adversary fixes a threshold f_t , generates ciphertexts offline (this is legal: the randomness is $H(\text{ID}(pk), m)$ for a chosen m), and keeps those with $p(c) > f_t$. With $\alpha = \Pr_c[p(c) > f_t]$ and $\beta = \mathbb{E}[p(c) \mid p(c) > f_t]$, the expected cost of one failure is α^{-1} encapsulations offline and β^{-1} decapsulation queries, i.e. total classical work $(\alpha\beta)^{-1}$; Grover search reduces the offline part to $\alpha^{-1/2}$. Since

$$\alpha\beta = \mathbb{E}[p(c) \mathbf{1}_{p(c) > f_t}] \leq \mathbb{E}[p(c)] = \delta, \quad (3)$$

classical total work never drops below $1/\delta$; boosting trades oracle queries for offline work. When the oracle is capped at 2^{64} queries (the NIST model), f_t is chosen so that $\beta^{-1} = 2^{64}$. Because $\text{ID}(pk)$ enters the hash, weak ciphertexts do not transfer between keys and the multi-target improvements of [4] do not apply.

3 The failure region of Algorithm 2

Lemma 1. *Let $u \in \{0, 1\}^8$ be the sent codeword and let all coefficients of $\text{Err}^{(i)}$ be represented in $(-q/2, q/2)$. Algorithm 2 returns a codeword $u' \neq u$ only if, with $D = \text{supp}(u' - u) \cup (\text{supp}(u' - u) + 8)$,*

$$\sum_{j \in D} |\text{Err}_j^{(i)}| \geq |D| \cdot \frac{q+1}{4}, \quad (4)$$

and conversely it fails whenever (4) holds for some codeword $u' \neq u$ (up to rounding of ± 1 in the threshold).

Proof. Let $h = (q+1)/2$. On a coordinate where $u_j \neq u'_j$, the received value is $hu_j + e_j$ and the two candidate symbols are hu_j and $hu_j + h \equiv hu_j - (q-1)/2$. The centred distance to the wrong symbol is $h - |e_j|$ (up to one unit) regardless of the sign of e_j , because 0 and h are antipodal in $\mathbb{Z}_q$. The cost difference of u' against u is therefore $\sum_{j \in D} ((h - |e_j|)^2 - e_j^2) = \sum_{j \in D} (h^2 - 2h|e_j|)$, which is negative iff $\sum_{j \in D} |e_j| > |D|h/2$. The codeword pair is repeated on both octets, so D contains a position and its partner at distance 8 inside the block. $\square$

The $[8, 4, 4]$ code has 14 codewords of weight 4 and one of weight 8, so a block has 14 half-space families with $|D| = 8$ (threshold $2(q+1)$) and one with $|D| = 16$. Writing (4) as $\max_{s \in \{\pm 1\}^D} \langle s, \text{Err}^{(i)} \rangle \geq |D|(q+1)/4$, the failure region is the union over $14 \cdot 2^8 + 2^{16}$ half-spaces, each at Euclidean distance $(q+1)/\sqrt{2}$ from the origin. The ball (2) has radius $(q-1)/\sqrt{2}$: it is the inscribed ball of the Voronoi cell and a strict subset of the correct-decoding region. We verified Lemma 1 against the reference decoder on 8000 random error vectors at $q \in \{193, 3457\}$: the predicate never fired on a correctly decoded block and missed only a handful of failures at the modular seam.

A *fixed-sign* variant that only keeps $s = u' - u$ misses almost all failures (2–4 of several hundred in our tests): because of the antipodal symbols, both signs contribute. This matters below, since the fixed-sign direction is the low-variance direction of the block.

4 Block covariance in the tricyclotomic ring

Fix the ciphertext randomness (r, u, w) and set $v = 2u + w$ (tricyclotomic and LPPNF) or $v = (1 - x^{n/2})u + w'$ (power-of-two). By (1), $\text{Err} = M_r g + M_v f' + u$ is linear in the secret (g, f') , whose coefficients are i.i.d. centred binomial with variances $\sigma_g^2 = k_g/2$ and $\sigma_{f'}^2 = k_{f'}/2$. For a block with index set I ,

$$\Sigma_I = \sigma_g^2 M_r[I] M_r[I]^\top + \sigma_{f'}^2 M_v[I] M_v[I]^\top = \sigma_g^2 \text{Gram}_I(r) + \sigma_{f'}^2 \text{Gram}_I(v). \quad (5)$$

The specification's model corresponds to replacing Σ_I by its diagonal (in fact by a single scalar).

Lemma 2. *In $\mathbb{Z}[x]/(x^n - x^{n/2} + 1)$, for every $a \in \mathcal{R}$ and every output index k , each coefficient a_j appears in $(a \cdot x^t)_k$ for exactly one t except for pairs (j, j') with $|j - j'| = n/2$, which appear together with equal sign. Consequently $\text{Gram}(a)_{k, k+n/2} \neq 0$ in general, while for $|k - k'| \notin \{0, n/2\}$ the entry $\text{Gram}(a)_{k, k'}$ has mean zero over random a .*

The proof is a direct computation with $x^n = x^{n/2} - 1$; in the power-of-two ring $x^n = -1$ produces no such pairing, and in the LPPNF ring $x^n = x + 1$ produces an analogous but weaker coupling. The consequence for DTRU is that a decoding block consists of exactly the eight ring-paired index pairs $(8i + j, 8i + j + n/2)$, $j = 0, \dots, 7$, and the encoding repetition $w_j = w_{j+n/2}$ adds a second correlated component through $\text{Gram}(v)$.

Numerical picture (DTRU-2048). Sampling the secret 3000 times for one honest ciphertext, coefficients 0 and $n/2$ of Err have variances $1.09 \cdot 10^4$ and $1.13 \cdot 10^4$ and covariance $-7.3 \cdot 10^3$ (correlation -0.66). The expected block covariance of block 0 has eigenvalues

$$\begin{aligned} &\{5709, 5723, 5781, 5788, 5797, 5843, 5851, 6767\} \cup \\ &\{28964, 28993, 29110, 29394, 29537, 29557, 29580, 38418\}, \end{aligned}$$

against a marginal variance of about $1.84 \cdot 10^4$. The trace is preserved: correlation redistributes variance, it does not create it. Since the failure region is a union of half-spaces, the tail is governed by $\max_s s^\top \Sigma_I s$, i.e. by the large eigenvalues, and the sign-alternating patterns s with $s_j = -s_{j+8}$ are exactly the ones aligned with them. Across the parameter sets the ratio between the largest eigenvalue and the mean diagonal is 2.1–2.5 (tricyclotomic), 3.4 (LPPNF) and 1.27 (power-of-two, DTRU-Light), see Table 3.

5 Estimating the failure probability

Per-ciphertext probability. Given (r, u, v) , for each block I and each pattern (D, s) we compute $\sigma_s^2 = s^\top \Sigma_I s$ and the Gaussian tail $Q(|D|(q+1)/4 \mp \langle s, u_I \rangle) / \sigma_s$ for both signs of s (the known term u shifts the mean); the block probability is the union bound over patterns, and $p(c)$ the union over blocks. At the operating points of DTRU the dominant pattern has standard score $z \approx 12$ –13, the half-spaces barely overlap, and the union bound is tight; on the toy parameters below (with $z \approx 3$ –4) it overestimates the mean by 1–2 bits, which we report.

Exact secret distribution. The Gaussian approximation of $\langle s, \text{Err} \rangle = a_g^\top g + a_{f'}^\top f'$ at 12σ deserves a check. The exact cumulant generating function is $K(t) = \sum_j 2k_g \log \cosh(ta_{g,j}/2) + \sum_j 2k_{f'} \log \cosh(ta_{f',j}/2)$, and the Lugannani–Rice saddlepoint formula gives the tail. Applied to the 64 most probable (I, D, s) terms of each ciphertext, it lowers $\log_2 p$ by 0.4–1.3 bits; all headline numbers include this correction.

Ring	toy parameters	Alg. 2 fails	ball	model median	model mean	$\mathbb{E}[\text{Err}^2]$ ratio
$x^n - x^{n/2} + 1$	$n=64, q=193, q_2=64, \mathcal{B}_{5,1,2,1}$	$2^{-9.7}$ (24)	$2^{-4.7}$	$2^{-10.0}$	$2^{-8.1}$	1.03
$x^n + 1, p=1-x^{32}$	$n=64, q=89, q_2=32, \mathcal{B}_{2,1,1,2}$	$2^{-8.1}$ (71)	$2^{-2.2}$	$2^{-8.4}$	$2^{-6.9}$	1.02
$x^n - x - 1$	$n=61, q=149, q_2=64, \mathcal{B}_{2,1,2,1}$	$2^{-8.4}$ (58)	$2^{-3.8}$	$2^{-9.2}$	$2^{-6.6}$	1.01

Table 2: Real decoder versus models on scaled-down instances (20000 encryptions each). “ball” is the rate of $\|\text{Err}^{(i)}\|^2 \geq T$; the last column is the empirical second moment of the error coefficients over the model’s $\text{tr } \Sigma_I/16$.

Set	claimed	indep. χ^2 + ball	σ^2 ours/ implied	spectrum + ball, mean	median	true decoding failure mean [95%]	normal tail	$\lambda_{\max}/$ diag
Light	-143.9	-139.2	1177/1145	-101.4	-153.6	-137.4 [-139.9, -136.2]	-121.7	1.27
648	-146.8	-153.3	21978/22777	-70.4	-96.8	-85.4 [-89.9, -84.1]	-83.6	2.13
768	-185.6	-189.7	18377/18719	-74.9	-111.6	-97.9 [-100.5, -96.6]	-86.0	2.31
1024	-190.5	-196.7	17783/18276	-83.9	-117.6	-104.6 [-108.2, -103.4]	-98.1	2.25
Prime	-180.7	-224.7	5402/6498	-64.4	-115.2	-96.3 [-102.0, -94.7]	-79.8	3.43
1536	-195.5	-195.6	17826/17834	-77.3	-110.7	-100.7 [-102.9, -99.4]	-93.9	2.50
2048	-204.1	-209.6	16781/17166	-88.6	-123.2	-110.8 [-115.2, -109.3]	-109.1	2.32

Table 3: $\log_2$ of failure probabilities, 200 honest ciphertexts per set. “implied” is the per-coefficient variance that reproduces the claimed DFR under the specification’s model.

Averaging over ciphertexts. For each set we draw 200 honest ciphertexts and report the median of $\log_2 p(c)$, the sample mean $\bar{p}$ of $p(c)$ (with a bootstrap interval), and the normal-tail extrapolation $\log_2 \mathbb{E}[p] = \mu + \sigma^2 \ln 2/2$ from the sample mean μ and standard deviation σ of $\log_2 p$. The sample mean is dominated by the few weakest ciphertexts; the extrapolation places the mass of $\mathbb{E}[p]$ at $\mu + \sigma^2 \ln 2$, i.e. 4–7 σ above the mean, beyond the sample. Within the sample, $\log_2 p$ is close to normal (skewness -0.23 to +0.04, excess kurtosis about 0). Since $\log_2 p$ is a concave function of the top eigenvalue, a Gaussian-tailed eigenvalue would make the true right tail lighter than normal, so the truth is expected between the two estimates.

5.1 Validation on scaled-down instances

For each ring we ran the complete scheme (key generation with polynomial inversion, compression, decompression, Algorithm 2) on 40 fresh keys with 500 encryptions each, keeping the CBD widths of a real parameter set and lowering (n, q) until failures become observable. Table 2 compares the observed failure rate with the ball condition (2) and with the model of this section evaluated on the same ciphertexts. The algebraic error (1) matched the integer computation of $c'f$ coefficient by coefficient in all 60000 encryptions.

The ball condition fires 4–6 bits more often than the decoder fails. The per-ciphertext model matches the decoder to 0.3–0.8 bits at the median; its mean is inflated by the union bound at these moderate standard scores. Per-key failure counts ranged from 0 to 21, so the failure rate of a fixed key deviates from the average by a few bits.

6 Corrected failure rates

Table 3 gives, for each set: the claimed DFR; the specification’s method (independent $\chi^2(16)$ on the ball) re-run with our marginal variance; the ball event with the correct spectrum; and the true decoding failure with the correct spectrum (median, sample mean with a 95% bootstrap interval, normal-tail extrapolation).

Three observations.

1. The third column reproduces the specification’s figures to within ± 8 bits for all sets

Set	level	Core-SVP ours/claimed	$\log_2$ work (sample mean)	(normal tail)	verdict (DFR)
Light	128	132.3 / 132	137.4	121.7	borderline
648	128	164.7 / 164	85.4	83.6	−43 bits
768	192	199.4 / 195	97.9	86.0	−94 to −106
1024	256	277.7 / 270	104.6	98.1	−151 to −158
Prime	256	292.6 / 280	96.3	79.8	−160 to −176
1536	384	416.7 / 411	100.7	93.9	−283 to −290
2048	512	512.5 / 567	110.8	109.1	−401

Table 4: Classical work for one decryption failure (decapsulations), without a query cap. Core-SVP values are for the NTRU key-recovery problem; the DTRU-2048 estimate is a primal-BDD fallback with low confidence.

except DTRU-Prime, where the specification’s implied variance is 20% larger than ours (its independent model would give 2^{-225} rather than the claimed 2^{-181}). The discrepancy between claimed and true DFR is therefore a modelling error, not a variance error.

2. For the six sets over the tricyclotomic and LPPNF rings the true failure probability is 50–110 bits larger than claimed. The independence assumption alone accounts for about 100 bits (the true decoding event under $\Sigma_I = \lambda I$ would be 2^{-221} for DTRU-2048); the ball over-approximation goes the other way by 18–50 bits. The specification’s 2^{20} allowance for adversarial messages does not cover this.
3. DTRU-Light over $x^n + 1$ has no ring pairing ($\lambda_{\max}$ is only $1.27\times$ the diagonal) and its median DFR is below the claim. Its spread across ciphertexts is however the largest ($\sigma = 9.6$ bits at $n = 512$), and the two estimates of $\mathbb{E}[p]$, 2^{-137} and 2^{-122} , straddle the 128-bit level.

7 Failure boosting against DTRU

7.1 Cost of the first failure

By (3) the classical total work for one failure is at least $1/\mathbb{E}[p]$. Table 4 lists it for the sample-mean and the normal-tail estimates, together with the nominal level and the NTRU Core-SVP estimates (ours, from the lattice estimator, and the designers’).

Filtering within the sample (keeping the weakest quarter, or the single weakest ciphertext) leaves the classical total unchanged and saves 3–4 bits quantumly, as (3) predicts. We also tried tilting the distributions of r, e and of the codeword weight (exponential families on the expected covariance): on DTRU-2048 this gains only a few bits, because $\|r\|^2$ is concentrated at $n = 2048$ and the weak ciphertexts are those with a large top eigenvalue (correlation 0.99 between $\log_2 p$ and the top eigenvalue on the ball statistic), not those with a large norm.

7.2 Precomputation versus oracle queries

Following [2], the offline and online costs separate. We fit $\log_2 p$ by a normal law and ask for the threshold f_t at which the conditional failure probability is 2^{-80} , and then for the threshold at which $\beta^{-1} = 2^{64}$.

For DTRU-648, -768 and -Prime the first failure fits into a 2^{64} query budget after 2^{22} – 2^{26} offline encapsulations; for DTRU-1024 and -1536 the offline part is 2^{37} – 2^{41} . The total work remains 2^{80} – 2^{130} , so none of this is a practical attack; the point is that the designers’ query-limit argument (“ $2^{64} \cdot 2^{-147}$ is negligible”) does not hold once weak ciphertexts are selected offline.

Set	$f_t = 2^{-80}$			oracle capped at 2^{64}			
	$\#\{p > 2^{-80}\}/200$	precomp	oracle	precomp	oracle	classical	quantum
648	2	8.3	75.4	24.1	62.2	86.3	74.2
768	0	13.0	73.0	25.9	60.6	86.6	73.6
Prime	0	12.1	67.7	22.3	57.5	79.8	68.7
1024	0	21.8	77.1	41.1	62.4	103.4	82.9
1536	0	17.6	76.9	36.6	62.4	99.0	80.7
2048	0	37.4	78.7	67.3	63.1	130.4	96.8
Light	0	46.8	77.7	67.5	62.3	129.8	96.1

Table 5: Offline precomputation ($\log_2 \alpha^{-1}$ encapsulations) and oracle queries ($\log_2 \beta^{-1}$) under the normal-tail fit. For DTRU-648 the 2^{-80} threshold is inside the sample: two of 200 honest ciphertexts have $p > 2^{-80}$ (weakest $2^{-78.7}$). The 2^{64} -cap entries for DTRU-2048 and DTRU-Light are 9σ extrapolations.

8 After the first failure

A failing ciphertext with its most probable pattern (I, D, s) tells the adversary that $\langle s, \text{Err}_I \rangle = a_g^\top g + a_f^\top f' + \langle s, u_I \rangle$ exceeded $|D|(q+1)/4$, where $a_g = M_r[I]^\top s$ and $a_f = M_v[I]^\top s$ are known. Implicit rejection hides the block and the pattern, so the posterior is a mixture whose largest component has weight around 10%; the sign of $\langle a, x \rangle$ is not revealed either.

For DTRU-2048 (24 honest ciphertexts) the most probable pattern has standard score 12.97 (median), the direction a has participation ratio $(\sum a_i^2)^2 / \sum a_i^4 \approx 891$ out of 4096 coordinates, and directions of different ciphertexts have absolute cosine 0.19 (median) to 0.27 (max). Conditioning on a first failure along a_1 , a second ciphertext with direction a_2 at correlation ρ fails with standard score $z' = (z - \rho z) / \sqrt{1 - \rho^2}$, i.e. 10.7 ($\rho = 0.19$) to 9.8 ($\rho = 0.27$), so its failure probability is 2^{-86} to 2^{-72} instead of 2^{-121} . This is the mechanism of directional failure boosting [3]: after the first failure, further failures are cheap, and the total cost is that of the first one.

On the $n = 64$ toy instance we collected 150 real decoder failures for one key and estimated the secret up to sign by the top eigenvector of $\sum_k a_k a_k^\top$: the estimate has correlation 0.22 with the true secret (random directions: 0.07), with only $z \approx 3.5$ per failure. At the real parameters each failure carries $z \approx 13$. We did not run the final lattice-reduction step and do not claim a completed key recovery.

9 Discussion

What is wrong and how to fix it. The specification computes the per-coefficient variance correctly (we reproduce its numbers) but then treats the 16 coefficients of a decoding block as independent. In $x^n - x^{n/2} + 1$ this is false in a structured way: the block is made of ring-paired coefficients, and the repetition code places the same message bit on both members of each pair. The correct computation is cheap: form the 16×16 conditional covariance Σ_I per block and evaluate the half-space union of Lemma 1 (or, conservatively, the generalised χ^2 tail of the ball on the eigenvalues of Σ_I). At the design stage one can also remove the coincidence: if the two octets of a decoding block are not ring-paired (for instance by using a repetition offset different from $n/2$, or by interleaving block positions so that no block contains both j and $j + n/2$), the pairing correlation moves to different blocks where it does not affect the per-block tail. We have not evaluated the correctness or cost impact of such a change.

Security levels. In the DFR metric, six of seven DTRU sets miss their nominal level by 43–400 bits, and DTRU-648 does so with a first-failure cost of about 2^{85} that fits into a 2^{64} -query budget after 2^{24} offline encapsulations. The FO advantage term $(q_H + q_D)\delta$ of the specification’s Theorem 3 becomes constant at $q_D \approx 2^{85} - 2^{111}$ with the corrected δ . None of this yields a practical attack, and DTRU-Light is unaffected in structure.

Limitations. Our estimate of $\mathbb{E}[p]$ is a tail quantity: the sample mean of 200 ciphertexts and the normal-tail extrapolation differ by 1–2 bits for DTRU-648 and -2048 and by 12–16 bits for DTRU-768, -Prime and -Light. Narrowing this requires importance sampling of ciphertexts tilted towards large top eigenvalues, which we leave open. Failure probabilities are averaged over keys; weak keys were not studied. The layout of DoubleE₈ for DTRU-Prime is our assumption, and the specification’s 20% larger variance for that set is unexplained. The directional-boosting figures are single-hyperplane conditional estimates, not a re-implementation of [3], and no lattice reduction on a failure-augmented instance was run.

10 Conclusion

For lattice KEMs that decode blocks of coefficients with an error-correcting code, the DFR analysis must account for correlations inside a block. DTRU’s tricyclic and LPPNF instantiations pair the coefficients of every decoding block through the ring itself, and its repetition encoding reinforces the pairing. Correcting the model raises the failure rate of six parameter sets by 50–110 bits and places the cost of a first decryption failure between 2^{85} and 2^{111} decapsulations, below every nominal level from 128 to 512. The computation that exposes this uses only 16×16 covariance matrices and can be added to any DFR script.

Reproducibility. All scripts (`dfr_all_sets.py`, `ml_decoder_check.py`, `exact_cgf_check.py`, `recover.py`) and their JSON outputs are provided with this paper. They depend only on NumPy, SciPy and mpmath.

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